

Probability Density Function Estimation: Theory and Mathematics CPE 486/586

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Sept 14, 2025

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1 Introduction and Problem Formulation

1.1 Problem Statement

Given a time series of room temperature measurements $X_1, X_2, \dots, X_n$ collected every 5 minutes from 9 AM to 9 PM, we want to estimate the probability density function $f(x)$ that describes the distribution of temperature values.

1.2 Mathematical Framework

Let X be a continuous random variable representing room temperature. The probability density function $f(x)$ satisfies:

$$f(x) \geq 0 \text{ for all } x \quad (1)$$

$$\int_{-\infty}^{\infty} f(x) dx = 1 \quad (2)$$

$$P(a \leq X \leq b) = \int_a^b f(x) dx \quad (3)$$

Our goal is to estimate $f(x)$ from the observed sample $\{x_1, x_2, \dots, x_n\}$.

2 Data Structure and Sampling Considerations

2.1 Temporal Structure

With measurements every 5 minutes from 9 AM to 9 PM:

- Time points: $t_i = 9 : 00 + 5i$ minutes, $i = 0, 1, \dots, 143$
- Daily sample size: $n = 144$ observations
- For k days of data: $N = 144k$ total observations

2.2 Stationarity Assumptions

The basic PDF estimation assumes that observations are independent and identically distributed (i.i.d.). However, room temperature data may exhibit:

- **Temporal dependence:** $X(t)$ may be correlated with $X(t - 1)$
- **Diurnal patterns:** $f(x|t)$ may vary with time of day
- **Non-stationarity:** Distribution parameters may change over time

However, in the most simplest case, we will assume i.i.d.

3 Parametric PDF Estimation

3.1 Normal Distribution Model

If we assume $X \sim N(\mu, \sigma^2)$, the PDF is:

$$f(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (4)$$

3.1.1 Parameter Estimation using Maximum Likelihood

The likelihood function for n observations is:

$$L(\mu, \sigma^2) = \prod_{i=1}^n f(x_i; \mu, \sigma^2) \quad (5)$$

The log-likelihood is:

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \quad (6)$$

Maximum Likelihood Estimators:

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (7)$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (8)$$

3.2 Other Parametric Distributions

3.2.1 Beta Distribution (for bounded temperature ranges)

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} \text{ for } x \in [0, 1] \quad (9)$$

After rescaling temperature to $[0, 1]$ using: $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$

3.2.2 Gamma Distribution (for positive temperatures in Kelvin)

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} \text{ for } x > 0 \quad (10)$$

3.3 Goodness-of-Fit Testing

3.3.1 Kolmogorov-Smirnov Test

Test statistic:

$$D_n = \sup_x |F_n(x) - F_0(x)| \quad (11)$$

Where $F_n(x)$ is the empirical CDF and $F_0(x)$ is the theoretical CDF.

3.3.2 Anderson-Darling Test

$$A = -n - \sum_{i=1}^n \frac{2i-1}{n} \left[\ln F(x_{(i)}) + \ln \left(1 - F(x_{(n+1-i)}) \right) \right] \quad (12)$$

Source: <https://www.statsref.com/HTML/anderson-darling.html>

4 Non-Parametric PDF Estimation

4.1 Kernel Density Estimation (KDE)

The kernel density estimator is:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right) \quad (13)$$

Where:

- $K(u)$ is the kernel function
- h is the bandwidth parameter
- n is the sample size

Source: Chapter 4, Harrou, Fouzi, Abdelhafid Zeroual, Mohamad Mazen Hittawe, and Ying Sun. Road traffic modeling and management: Using statistical monitoring and deep learning. Elsevier, 2021.

4.1.1 Common Kernel Functions

Gaussian Kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) \quad (14)$$

Epanechnikov Kernel (optimal):

$$K(u) = \frac{3}{4}(1 - u^2) \text{ for } |u| \leq 1, 0 \text{ otherwise} \quad (15)$$

Uniform Kernel:

$$K(u) = \frac{1}{2} \text{ for } |u| \leq 1, 0 \text{ otherwise} \quad (16)$$

4.2 Bandwidth Selection

4.2.1 Silverman's Rule of Thumb

$$h = 0.9 \times \min(\hat{\sigma}, \text{IQR}/1.34) \times n^{-1/5} \quad (17)$$

Where IQR is the interquartile range.

4.2.2 Cross-Validation Bandwidth

Minimize the integrated squared error:

$$\text{ISE}(h) = \int [\hat{f}(x) - f(x)]^2 dx \quad (18)$$

Practical Cross-Validation:

$$h^* = \arg \min \sum_{i=1}^n \int [\hat{f}_{-i}(x) - \delta(x - x_i)]^2 dx \quad (19)$$

Where $\hat{f}_{-i}(x)$ is the KDE excluding observation x_i .

4.3 Histogram-Based Estimation

4.3.1 Basic Histogram

$$\hat{f}(x) = \frac{n_j}{n \times \Delta x} \text{ for } x \in [x_j, x_{j+1}) \quad (20)$$

Where n_j is the count in bin j and Δx is the bin width.

4.3.2 Optimal Bin Width (Sturges' Rule)

Number of bins: $k = \lceil 1 + \log_2(n) \rceil$

Bin width: $\Delta x = \frac{x_{\max} - x_{\min}}{k}$

4.3.3 Freedman-Diaconis Rule

$$\Delta x = 2 \times \text{IQR} \times n^{-1/3} \quad (21)$$

5 Time-Varying PDF Models

5.1 Conditional PDF Estimation

For non-stationary data, estimate $f(x|t)$:

5.1.1 Kernel Regression Approach

$$\hat{f}(x|t) = \sum_{i=1}^n w_i(t) K\left(\frac{x - x_i}{h}\right) \quad (22)$$

Where $w_i(t)$ are time-dependent weights:

$$w_i(t) = \frac{W\left(\frac{t-t_i}{b}\right)}{\sum_{j=1}^n W\left(\frac{t-t_j}{b}\right)} \quad (23)$$

5.2 Functional Data Analysis

Model temperature as a function $T(t)$ and estimate the distribution of functional parameters.

5.2.1 Fourier Representation

$$T(t) = \mu(t) + \sum_{k=1}^K [a_k \cos(2\pi kt/P) + b_k \sin(2\pi kt/P)] + \varepsilon(t) \quad (24)$$

Where $P = 12$ hours is the period.

6 Model Selection and Validation

6.1 Information Criteria

6.1.1 Akaike Information Criterion

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p \quad (25)$$

6.1.2 Bayesian Information Criterion

$$\text{BIC} = -2\ell(\hat{\theta}) + p \ln(n) \quad (26)$$

Where p is the number of parameters.

6.2 Cross-Validation

6.2.1 K-Fold Cross-Validation

1. Divide data into K folds
2. For each fold k , estimate $\hat{f}_{-k}(x)$ using remaining data
3. Evaluate log-likelihood on fold k : $\text{LL}_k = \sum_{i \in k} \ln(\hat{f}_{-k}(x_i))$
4. Average: $\text{CV-LL} = \frac{1}{K} \sum_{k=1}^K \text{LL}_k$

6.3 Bootstrap Confidence Intervals

6.3.1 Bootstrap Procedure

1. Resample with replacement: $X^* = \{x_1^*, x_2^*, \dots, x_n^*\}$
 2. Compute $\hat{f}^*(x)$
-

3. Repeat B times
4. Construct pointwise confidence intervals from bootstrap distribution

7 Practical Considerations

7.1 Sample Size Requirements

For reliable PDF estimation:

- Parametric: $n \geq 30$ typically sufficient
- Non-parametric KDE: $n \geq 100$ recommended
- Complex time-varying models: $n \geq 1000$ may be needed

7.2 Boundary Effects

Room temperatures are bounded (e.g., 15-30°C). Use:

- Boundary kernels for KDE
- Transformed variables
- Reflection methods

7.3 Computational Complexity

- Histogram: $O(n)$
- KDE: $O(n^2)$ for evaluation at n points
- Parametric MLE: $O(n)$ per iteration

8 Implementation Algorithm

8.1 General Workflow

Listing 1: General Workflow for PDF Estimation

```

1  1. Data Preprocessing:
2    - Remove outliers ( $|x - \text{median}| > 3 \times \text{MAD}$ )
3    - Check for temporal patterns
4    - Test for stationarity
5
6  2. Model Selection:
7    - Fit parametric candidates
8    - Compute non-parametric estimates
9    - Use cross-validation for comparison
10
11 3. Parameter Estimation:
12   - If parametric: MLE or method of moments
13   - If KDE: optimize bandwidth
14   - If histogram: optimize bin width
15
16 4. Validation:
17   - Goodness-of-fit tests
18   - Cross-validation
19   - Bootstrap confidence intervals
20
21 5. Final Estimation:
22   - Select best model
23   - Provide uncertainty quantification

```

8.2 Error Metrics

8.2.1 Mean Integrated Squared Error

$$\text{MISE} = E \left[\int (\hat{f}(x) - f(x))^2 dx \right] \quad (27)$$

8.2.2 Pointwise Mean Squared Error

$$\text{MSE}(x) = E[(\hat{f}(x) - f(x))^2] = \text{Bias}^2(x) + \text{Variance}(x) \quad (28)$$